12 October 2026

Dear Editor,

Please consider the enclosed manuscript, "Sensor degradation and maintenance-trigger errors in remaining-useful-life prediction intervals: a C-MAPSS stress test", as an Original Research Article for *Proceedings of the Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability*.

Prediction intervals on remaining useful life are increasingly used in maintenance rules that trigger work when a lower bound falls below a lead time. Interval quality is nevertheless judged mostly by coverage on clean inputs, which does not say which way a rule fails when the sensors feeding the model degrade. The paper studies that question in a controlled empirical stress test: five interval mechanisms on two backbones and the four C-MAPSS sub-datasets, with noise, bias, drift and gain perturbations of the sensor channels, and a one-sided lower-bound rule at three lead times whose premature and unrecognised triggers are counted separately.

The paper makes two empirical contributions. First, in the conditions tested, the side on which the lower bound errs follows the sign pattern of the sensor shift. A positive drift on all sensor channels triggered maintenance prematurely for 52–69% of evaluated six-condition engines at a 20-cycle lead time without any unrecognised need, whereas a negative constant offset produced unrecognised needs for 4–15%; coverage and the share of out-of-range inputs did not reveal this direction. Second, the trigger errors are reported with engine-level resampling intervals and a mean/scale output swap that identifies the mean shift as the main driver (with scale changes amplifying or offsetting it), and the cost ranking of mechanisms is shown to be unstable across cost ratios (36 of 96 settings, none under drift). We believe this is relevant to readers of the journal because it concerns how the reliability of condition-based maintenance decisions should be assessed when sensing degrades.

The evidence has clear limits, which the manuscript states: simulated data, two backbones, one terminal decision per engine, a binary cost and exploratory engine-level analyses; it does not support deployment safety margins, and the study makes no priority claim in the literature.

A previous version of this work, with a different title, was submitted to IEEE Transactions on Reliability and was returned on 9 October 2026 on grounds of fit to that journal; no scientific or technical criticism was given. The manuscript is not under consideration elsewhere. Code, checkpoints, result files and the scripts that generate the tables and numbers are public at https://github.com/jeff-1980/paper2-JRR. The AI assistant Claude (Anthropic) was used for analysis code, text editing and consistency checks; the author verified all outputs and is responsible for the content. The author declares no conflicting interests and no funding.

Yours sincerely,

Yi Wang
College of Engineering, Lishui University, Lishui 323000, Zhejiang, China
jefflsxy@lsu.edu.cn, +86 13757838533